

A Kernel-Checked Proof of Fuglede's Conjecture for the Cyclic Group $\mathbb{Z}/180\mathbb{Z}$

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Abstract. This article gives a machine-checked proof of both directions of Fuglede's conjecture for the cyclic group $\mathbb{Z}/180\mathbb{Z}$: a finite subset is spectral if and only if it tiles by translation. Spectrality is encoded exactly by cyclotomic divisibility of the integer mask polynomial, and tiling is bijectivity of the addition map. The spectral-to-tiling direction ends in a cardinality 30 classification. Its marginal audit compresses 16,796 literal pairs to 213 profile cells, refutes 16,574 pairs by exact integer coefficients, and links the remaining 222 to explicit catalogue witnesses; a common-frame certificate checks the coupled Gram data. For the converse, an exact Fourier zero-cover theorem reduces all tilings to their eighteen possible cardinalities. Structural closures, quotient descent, and a six-point theorem in $\mathbb{Z}/36\mathbb{Z}$ discharge those cases; the last thirty-point case is obtained by dilating its six-point complement, descending to $\mathbb{Z}/36\mathbb{Z}$, and lifting a six-frequency set through five fibres. Lean 4 and `leanchecker` replay the final biconditional with only `propext`, `Classical.choice`, and `Quot.sound`. The complete Lean dependency closure accompanies the article, together with the supplemental certificates, deterministic generators, and manifests used for the largest spectral-to-tiling branch.

1 Introduction

Fuglede's conjecture relates two rigid structures: a set that admits an orthogonal basis of characters and a set that tiles its ambient group by translations. In Euclidean spaces the unrestricted conjecture is false in high dimension, while important finite and one-dimensional cases retain a rich arithmetic structure [5, 13, 3]. Finite cyclic groups are especially useful because Fourier vanishing can be encoded by divisibility of integer mask polynomials by cyclotomic polynomials [1]. This turns analytic orthogonality into an exact, checkable statement about polynomials and roots of unity [8, 9].

Among the established cyclic families are groups of order $p^n q$, pqr , $p^2 qr$, and $pqrs$ [10, 11, 12, 6]; group-ring methods also treat $p^n qr$ [15]. A closer recent result proves the spectral-to-tiling direction for $\mathbb{Z}_{p^2 q^2 r}$ under the size condition $p^2 q^2 < r$ [4]. At order 180, however, this condition reads $36 < 5$ and therefore does not apply. The reverse direction is related to broader results on three-prime integer tilings and the Coven–Meyerowitz conditions [7]. The present work gives a self-contained kernel-checked proof specialized to order 180; it makes no claim about arbitrary moduli.

This article reports a complete formal proof of both directions for $G = \mathbb{Z}/180\mathbb{Z}$, where

$$180 = 2^2 \cdot 3^2 \cdot 5.$$

The spectral-to-tiling direction is driven by structural reductions and a finite classification. Its difficult residual case has $|A| = 30$ and requires both a marginal classification and compatibility of a five-row Gram configuration. The converse starts from the exact Fourier characterization of a tiling pair, resolves all possible divisor cardinalities, and closes its last thirty-point case by descent to $\mathbb{Z}/36\mathbb{Z}$.

Every finite computation reaches the kernel as an ordinary Lean declaration, rather than remaining an oracle

beside the proof. For the largest generated V97 sub-DAG, deterministic scripts emit those declarations and hash manifests fail closed if sources, catalogues, witness locations, or import order change. The rest of the theorem is likewise committed as checkable Lean source. No floating-point calculation enters the theorem.

Contribution

The formal development establishes the finite cyclic Fuglede conjecture at order 180.

Theorem 1 (Kernel-checked Fuglede theorem at order 180). *For every $A \subseteq \mathbb{Z}/180\mathbb{Z}$, the following conditions are equivalent:*

- (i) *A tiles $\mathbb{Z}/180\mathbb{Z}$ by translation;*
- (ii) *A admits an orthogonal basis of characters.*

Equivalently, there is a finite set B for which the addition map $A \times B \rightarrow \mathbb{Z}/180\mathbb{Z}$ is bijective if and only if there is a finite set L that is an exact cyclotomic spectrum for A .

In the repository, Theorem 1 is the declaration `Fuglede.z180_tiles_iff_spectral`.

2 The exact finite formulation

For $A \subseteq \mathbb{Z}/N\mathbb{Z}$, choose the standard representatives $a \in \{0, \dots, N-1\}$ and define the mask polynomial

$$m_A(X) = \sum_{a \in A} X^a \in \mathbb{Z}[X].$$

For $d \in \mathbb{Z}/N\mathbb{Z}$, let

$$\text{ord}_N(d) = \frac{N}{\gcd(N, d)}.$$

The formal predicate `CyclotomicZero` requires

$$\Phi_{\text{ord}_N(d)}(X) \mid m_A(X) \quad \text{in } \mathbb{Z}[X].$$

This is the exact algebraic certificate used for Fourier vanishing at frequency d .

Definition 2 (Exact cyclotomic spectrum). A finite set $L \subseteq \mathbb{Z}/N\mathbb{Z}$ is a cyclotomic spectrum for A if $A \neq \emptyset$, $|A| = |L|$, and for every two distinct $\ell_1, \ell_2 \in L$,

$$\Phi_{\text{ord}_N(\ell_1 - \ell_2)} \mid m_A.$$

This predicate is not a substitute for the usual analytic definition. Let χ_N be the standard injective additive character of $\mathbb{Z}/N\mathbb{Z}$ and define the unnormalised Fourier coefficient

$$\widehat{1}_A(d) = \sum_{a \in A} \chi_N(ad).$$

The formalization proves the exact equivalence

$$\Phi_{\text{ord}_N(d)} \mid m_A \iff \widehat{1}_A(d) = 0. \quad (1)$$

It follows that `CyclotomicSpectrum` is equivalent to the ordinary pairwise Fourier-orthogonality condition. Because $|A| = |L|$ and $A \neq \emptyset$, the Fourier evaluation matrix is square with nonzero, orthogonal columns; its characters therefore form an orthogonal basis on A . Lean also proves the dual spectrum statement by converting column orthogonality into row orthogonality. Thus Theorem 1 assumes the standard finite notion of spectrality, expressed in an exact algebraic form.

The tiling predicate is deliberately independent of Fourier analysis. For finite sets $A, B \subseteq G$, let

$$f_{A,B} : A \times B \longrightarrow G, \quad f_{A,B}(a, b) = a + b.$$

Lean defines

$$\text{Tiles}(A, B) \iff f_{A,B} \text{ is bijective.}$$

Thus the conclusion includes both existence and uniqueness of each representation. In particular, it implies $|A||B| = |G|$.

In ASCII notation, the public endpoint has the following type; the two imported endpoints supply its proof.

```
theorem z180_tiles_iff_spectral
  (A : Finset (ZMod 180)) :
  Iff
    (Exists fun B : Finset (ZMod 180) => Tiles A B)
    (Exists fun L : Finset (ZMod 180) =>
      CyclotomicSpectrum 180 A L)
```

The two imported endpoints contain the substantive work. The spectral-to-tiling classification is described first, followed by the tiling-to-spectral reduction.

3 The spectral-to-tiling direction

Figure 1 shows the dependency structure of the computationally harder implication. The global reduction isolates the cardinality-30 obstruction. That obstruction has a projective/marginal component (V97) and a coupled common-frame component (V95). V81 turns their catalogue into the global theorem, and V96 supplies the two certificates unconditionally.

The version labels in Figure 1 are immutable module suffixes: V97 is the projective-profile audit, V95 the common-frame certificate, V83 their catalogue interface, V81 the global reduction, and V96 the unconditional spectral-to-tiling endpoint.

3.1 The global reduction

The surrounding library converts the spectrum hypothesis into exact cyclotomic zero data, applies cardinality and divisor sieves, and closes the nonexceptional branches. Sets of size greater than 90 are handled by the upper-half reduction. At size at most 90, projection to complementary factors restricts the cardinality to a divisor of 180 or to

$$7, 8, 11, 13, 14, 16, 17, 19, 21, 24, 27, 33.$$

Kernel-checked nondivisor arguments eliminate all entries in this list except the intermediate 24 and 33 branches. The divisor reduction and subsequent closures discharge the residual sizes 6, 10, 12, 18; separate arguments close 33 and exclude 24. The master conditional theorem consequently asks only for a closure at cardinality 30. The module `Z180K30CatalogueMasterClosureV81` packages this interface: a complete exceptional catalogue implies the spectral-to-tiling theorem for all cardinalities.

This separation is valuable both mathematically and formally. The general Fourier and tiling lemmas do not depend on the large finite catalogue, while the catalogue proof sees a small and rigid normalized state space.

3.2 Projective normalization and ordered differences

The exceptional reduction starts from balanced cardinality-30 data and passes to five six-point row fibres and one six-point column fibre in $\mathbb{Z}/36\mathbb{Z}$. Fixing a column fibre gives five 6×6 Fourier blocks. Let V denote its six residues and $U_0, \dots, U_4$ the five row patterns; a marginal check below uses one such row and abbreviates it by U . The projective-cover alternative orients the pair so that V is the side whose differences have a nontrivial common divisor

$$d = \gcd(36, \{v - v' : v, v' \in V\}).$$

Validity of the six-point sets bounds d by 6, while the two-class condition on U eliminates $d = 1, 2$. Lean therefore proves $d \in \{3, 4, 6\}$. Thus U , V , and d below are normalized fibre data forced by the exceptional branch, not new assumptions.

The projective branch works with six-element subsets of $\mathbb{Z}/36\mathbb{Z}$. For a list S , define its ordered-difference list

$$\Delta(S) = ((x - y) \bmod 36)_{(x,y) \in S^2}.$$

Diagonal terms are retained. The scalar Gram coefficient used by the audit can be written as the bilinear sum

$$\Gamma(U, V) = \sum_{a \in \Delta(U)} \sum_{b \in \Delta(V)} \zeta_0(ab), \quad (2)$$

where ζ_0 is the exact integer-valued coefficient extracted from the cyclotomic representation. Lean proves Equation (2) by unfolding the Gram trace. Exchanging the two

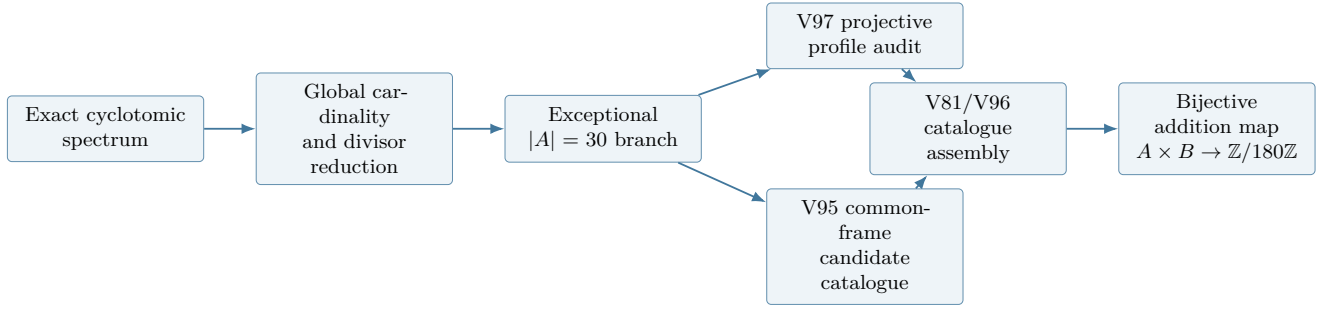


Figure 1: High-level structure of the checked proof. V97 and V95 are independent finite certificates joined only at catalogue assembly.

finite sums and using $ab = ba$ proves

$$\Gamma(U, V) = \Gamma(V, U).$$

This symmetry removes a duplicated orientation from the finite audit.

For a supported divisor $d \in \{3, 4, 6\}$, every difference on the normalized V -side is a multiple of d . The U -profile stores the responses

$$K_{d,U}(t) = \sum_{a \in \Delta(U)} \zeta_0(at), \quad 0 \leq t < 36,$$

with zero outside the multiples of d . The V -profile is the sorted list $\Delta(V)$, which is a canonical representation of its histogram. A proved compression lemma states

$$\Gamma(U, V) = \sum_{b \in \Delta(V)} K_{d,U}(b). \quad (3)$$

Sorting changes no sum because Lean carries an explicit permutation proof. The use of a sorted list rather than an unverified hash is important: profile equality is itself checked in the kernel.

3.3 The finite profile census

The normalized literal universe contains 16,796 distinct pairs. Grouping by the two exact profiles produces the much smaller table in Table 1.

Table 1: Authenticated projective-profile census.

d	U profiles	V profiles	cells	positive pairs
3	5	35	175	42
4	4	7	28	0
6	10	1	10	180
Total	19	43	213	222

The exceptional spectral argument determines the six eigenvalues of each Gram block: one is 30, one is 6, and four are zero. Consequently its exact trace-square target is

$$\text{tr}(G^2) = 30^2 + 6^2 = 936.$$

Each profile cell evaluates the corresponding integer coefficient. A value different from 936 refutes every pair in that

cell. This eliminates 16,574 pairs. A cell attaining 936 is not accepted merely because the total count is plausible: each of the remaining 222 literal pairs carries an exact pointer into one of the certified V87 witness shards. Lean checks the shard index, witness index, divisor, normalization, and the literal U, V values. The surviving counts $42 + 180$ therefore express a set equality, not a probabilistic or numerical match.

The profile compression was also essential for proof engineering. Expanding the original pairwise Gram computation retained millions of repeated root-of-unity reductions and exceeded practical memory limits. Equations (2)–(3) move the repeated work into a small set of reusable profiles without changing the certified proposition.

3.4 The common-frame branch

The marginal catalogue is insufficient by itself: five row configurations must be compatible with the same column frame. The V95 branch handles this coupled condition.

For a fixed normalized column orbit o , let $c_{e,k}(W, V_o)$ denote the exact integer coefficient at Gram entry e and cyclotomic coordinate k . The row signature is a short list

$$\sigma_o(W) = (c_{e_j, k_j}(W, V_o))_{j=1}^t, \quad t \in \{0, 1, 2\},$$

where the selected entries depend only on o . The coupled star condition retains the five signatures $(\sigma_o(U_0), \dots, \sigma_o(U_4))$ in one affine frame and requires their selected coordinates to sum to zero. It also records that each pair (U_i, V_o) has trace-square 936 and that the five Gram matrices sum to the diagonal target with value 30.

First, an orbit-frame extractor writes the normalized column as one affine image with a unit and a translation. The same unit is then applied to every row. Covariance lemmas transport the selected coordinates of all five rows into this common frame. Second, a trace classifier identifies the only compatible row orbit. Finally, a bounded affine table supplies a valid candidate with the same exact star signature. Permutation invariance transports the candidate from the representative to the original row.

The V83 catalogue interface returns both the five candidates and proofs that their exact coordinates realize the required star. The finite affine leaves, their aggregate, and the final common-frame module are all replayed separately.

Table 2: Cardinality closures for tiling-to-spectrality.

Cardinalities	Main mechanism
1, 2, 3, 180	Endpoint and direct character arguments
4, 5, 9	Prime-power allocation
60, 90	Small-complement Fourier construction
20, 36, 45	Exact cyclotomic forcing
10, 12, 15, 18	Quotient descent and lifted spectra
6, 30	Six-point forcing in $\mathbb{Z}/36\mathbb{Z}$

3.5 Catalogue assembly

The V97 normalization gives marginal coverage. The V95 framed-star object gives the coupled witness. V83 combines them into complete exceptional catalogue data; V81 converts that data into the global cardinality-30 closure and then invokes the previously completed cardinality reduction. V96 is the final three-import wrapper joining these components.

4 The tiling-to-spectral direction

The converse uses the same exact Fourier language but a different proof architecture. For a tiling pair (A, B) in a finite cyclic group, convolution and character orthogonality give

$$\begin{aligned} \text{Tiles}(A, B) \iff |A||B| = N \text{ and} \\ \forall d \neq 0, \quad (\widehat{\mathbf{1}}_A(d) = 0 \\ \text{or } \widehat{\mathbf{1}}_B(d) = 0). \end{aligned} \quad (4)$$

Lean proves both implications, including Fourier inversion from the right-hand side back to uniqueness of representations. Equation (1) then converts this zero cover to cyclotomic divisibility.

The cardinality identity in Equation (4) leaves exactly the eighteen positive divisors of 180. The reduction closes them in stages, with the final two cases isolated as $|A| = 6$ and $|A| = 30$. Table 2 summarizes the mechanisms used for all cases.

The six-point theorem in $\mathbb{Z}/36\mathbb{Z}$ is the reusable core. For a 6×6 tiling, the zero cover allocates exactly one of the orders 2, 4 and one of the orders 3, 9 to each factor. Previously checked forcing lemmas turn the remaining orders into one of five explicit difference patterns, each of which supplies a six-frequency spectrum. A six-point tile in $\mathbb{Z}/180\mathbb{Z}$ descends injectively modulo 36 and its spectrum lifts by multiplying frequencies by five.

For the last case, suppose $|A| = 30$. Its tiling complement has six elements. Multiplication by five preserves that complement as a tiling factor, while reduction modulo 36 sends it injectively to a six-point set C . An explicit six-frequency set $Q \subseteq \mathbb{Z}/36\mathbb{Z}$ is chosen so that the Fourier transform of C is nonzero on every nonzero difference from Q . The full preimage L of Q in $\mathbb{Z}/180\mathbb{Z}$ has $5 \cdot 6 = 30$ elements. The zero-cover property forces the Fourier transform of A to vanish on every nonzero difference in L , so L is a spectrum for A .

These constructions form the V12 tiling-to-spectral endpoint. Combining it with the V96 spectral-to-tiling endpoint yields the public theorem `z180_tiles_iff_spectral` without any additional finite search.

5 Formal verification and trust

The development uses Lean 4.31.0 and a pinned mathlib revision. Lean and mathlib provide a small trusted kernel and a large reusable mathematical library [2, 14]. The release adopts the following policy.

Exact arithmetic. Mask polynomials have integer coefficients. Finite Gram computations reduce to integer lists and exact cyclotomic coefficients. There is no floating-point tolerance and no numerical root approximation.

Untrusted generation, trusted checking. Python and PowerShell generate literal definitions, theorem statements, and replay schedules. They are convenience tools, not part of the logical trust base. Every generated declaration is elaborated by Lean. Structural roots are replayed again by `leanchecker`.

No proof escape hatches. The complete 2,411-module dependency closure of the final biconditional contains no `sorry`, `admit`, project-defined `axiom`, `native_decide`, or `unsafe`. In the large V97 sub-DAG, ordinary decisions are deliberately kept small; larger statements are split into bounded leaves and composed with proved list and permutation lemmas.

Fail-closed provenance. The master manifest authenticates 946 modules in a unique topological order, including an external algebraic closure of 757 modules. It pins child manifests, generators, catalogue inputs, the order of 66 witness shards, and the 16,796/213/222 census. Source drift causes authentication to stop before Lean is launched.

Axiom report. For both catalogue completeness and the final theorem, Lean prints exactly

[propext, Classical.choice, Quot.sound].

These are the standard classical and quotient principles used by mathlib; no problem-specific axiom appears.

6 Reproducing the artifact

The repository contains the exact transitive Lean source closure of the final theorem, the pinned toolchain, the article, and supplementary provenance for the large V97 certificate. The public artifact is available at <https://github.com/asim-v/fuglede-z180-formal>. The shortest kernel replay is:

```
cd fuglede_lean
lake update
lake exe cache get
lake build '+Fuglede.Z180FugledeTheorem:olean'
lake env leanchecker -v Fuglede.Z180FugledeTheorem
cd ..
```

Every proof-relevant finite certificate is an ordinary Lean source file in that import closure; no external solver or Python program is trusted by the final theorem. As an additional provenance check for the largest generated sub-DAG, run:

```
python scripts/generate_z180_k30_projective_profile_audit_v97.py
```

This command authenticates the V97 sources and manifest. The serialized PowerShell runner provides a module-by-module replay under a fixed memory cap.

7 Scope and interpretation

Theorem 1 is intentionally precise.

- It proves both spectral-to-tiling and tiling-to-spectral for $\mathbb{Z}/180\mathbb{Z}$.
- Equation (1) identifies the exact cyclotomic predicate with the usual finite Fourier definition; no approximate vanishing is assumed.
- Tiling means unique representation under addition, not merely a cardinality or covering condition.
- The theorem concerns one finite cyclic group. It is not a theorem for arbitrary moduli, arbitrary finite abelian groups, or Euclidean spaces.
- The large finite leaves are proof terms checked by Lean and are best inspected and replayed as code.

This distinction is more than cautious wording. Formal verification certifies that the encoded theorem follows from its definitions; it does not replace the mathematical task of stating those definitions faithfully. For that reason the exact Lean predicates and the public endpoint are included explicitly.

8 Conclusion

Fuglede’s conjecture holds for the finite cyclic group $\mathbb{Z}/180\mathbb{Z}$. The two directions require complementary ideas. Spectral-to-tiling uses a structured finite certificate whose difficult cardinality-30 branch separates marginal profile data from coupled common-frame data. Ordered-difference profiles compress its marginal audit to 213 exact cells, and explicit witness pointers prevent positive cases from being accepted by count alone. Tiling-to-spectral uses an exact Fourier zero cover, cardinality reduction, and quotient descent; its final case lifts a six-frequency construction from $\mathbb{Z}/36\mathbb{Z}$.

The final declaration merely joins these independently checked endpoints, but its complete transitive source closure is public: definitions, algebraic identities, finite tables, generated proof terms, hashes, and replay commands are all exposed. The result is therefore both a theorem about a concrete cyclic group and a reproducible case study in turning large finite harmonic-analysis arguments into kernel-checked mathematics.

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